

Solution to Exercise 4.13 (Page 84)

Problem 1)

- $$\begin{cases} \min_{\mathbf{x} \in \mathbb{R}^2} & \frac{1}{2}(x_1^2 + x_2^2) \\ \text{s.t.} & x_1 = 1 \end{cases}$$

- Optimal solution: $\mathbf{x}^* = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ with $\lambda^* = -1$

- Augmented Lagrangian

$$L_c(\mathbf{x}, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) + \lambda(x_1 - 1) + \frac{c}{2}(x_1 - 1)^2$$

- Minimization step $\mathbf{x}^k = \arg \min_{\mathbf{x}} L_{c^k}(\mathbf{x}, \lambda^k)$:

$$\nabla_{\mathbf{x}} L_{c^k}(\mathbf{x}, \lambda^k) = \begin{bmatrix} x_1 + \lambda + c^k(x_1 - 1) \\ x_2 \end{bmatrix} \stackrel{!}{=} \mathbf{0} \Rightarrow \mathbf{x}^k = \begin{bmatrix} \frac{c^k - \lambda^k}{c^k + 1} \\ 0 \end{bmatrix}$$

- Multiplier update

$$\lambda^{k+1} = \lambda^k + c^k(x_1^k - 1) = \dots = \frac{\lambda^k - c^k}{c^k + 1}$$

- Reformulation with $\lambda^* = -1$:

$$\begin{aligned} \lambda^{k+1} - \lambda^* &= \frac{\lambda^k - c^k}{c^k + 1} - \lambda^* \\ &= \frac{\lambda^k - \lambda^*}{c^k + 1} - \underbrace{(1 + \lambda^*)}_{=0} \frac{c^k}{c^k + 1} = \frac{\lambda^k - \lambda^*}{c^k + 1} \end{aligned}$$

- Effect of penalty parameter c^k

- Multiplier “dynamics” $\Delta \lambda^{k+1} = \frac{1}{c^k + 1} \Delta \lambda^k$ with $\Delta \lambda^k = \lambda^k - \lambda^*$

- Asymptotic stability for $-1 < \frac{1}{c^k + 1} < 1$

- $\lambda^k \rightarrow \lambda^*$ and $\mathbf{x}^k \rightarrow \mathbf{x}^*$ for every sequence $\{c^k\}$ with $c^k \geq c^{k-1} \geq \bar{c} > 0$

Problem 2)

- $$\begin{cases} \min_{\mathbf{x} \in \mathbb{R}^2} & \frac{1}{2}(-x_1^2 + x_2^2) \\ \text{s.t.} & x_1 = 1 \end{cases}$$

- Optimal solution: $\mathbf{x}^* = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ with $\lambda^* = 1$

- Augmented Lagrangian

$$L_c(\mathbf{x}, \lambda) = \frac{1}{2}(-x_1^2 + x_2^2) + \lambda(x_1 - 1) + \frac{c}{2}(x_1 - 1)^2$$

- Minimization step $\mathbf{x}^k = \arg \min_{\mathbf{x}} L_{c^k}(\mathbf{x}, \lambda^k)$:

$$\nabla_{\mathbf{x}} L_{c^k}(\mathbf{x}, \lambda^k) = \begin{bmatrix} -x_1 + \lambda + c^k(x_1 - 1) \\ x_2 \end{bmatrix} \stackrel{!}{=} \mathbf{0} \Rightarrow \mathbf{x}^k = \begin{bmatrix} \frac{c^k - \lambda^k}{c^k - 1} \\ 0 \end{bmatrix}$$

- Multiplier update

$$\lambda^{k+1} = \lambda^k + c^k(x_1^k - 1) = \dots = \frac{c^k - \lambda^k}{c^k - 1}$$

- Reformulation with $\lambda^* = 1$:

$$\begin{aligned} \lambda^{k+1} - \lambda^* &= \frac{c^k - \lambda^k}{c^k - 1} - \lambda^* \\ &= -\frac{\lambda^k - \lambda^*}{c^k - 1} + \underbrace{(1 - \lambda^*)}_{=0} \frac{c^k}{c^k - 1} \\ &= -\frac{\lambda^k - \lambda^*}{c^k - 1} \end{aligned}$$

- Effect of penalty parameter c^k

- Uniqueness of $\mathbf{x}^k$ requires $c^k \neq 1$ ($c^k > 1$)

- Multiplier “dynamics” $\Delta \lambda^{k+1} = -\frac{1}{c^k - 1} \Delta \lambda^k$ with $\Delta \lambda^k = \lambda^k - \lambda^*$

- Asymptotic stability for $-1 < -\frac{1}{c^k - 1} < 1$

- $\lambda^k \rightarrow \lambda^*$ and $\mathbf{x}^k \rightarrow \mathbf{x}^*$ for every sequence $\{c^k\}$ with $c^k \geq c^{k-1} \geq \bar{c} > 2$
(because of non-convexity of the problem)